Weekly Meeting
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μLHC Physics Update

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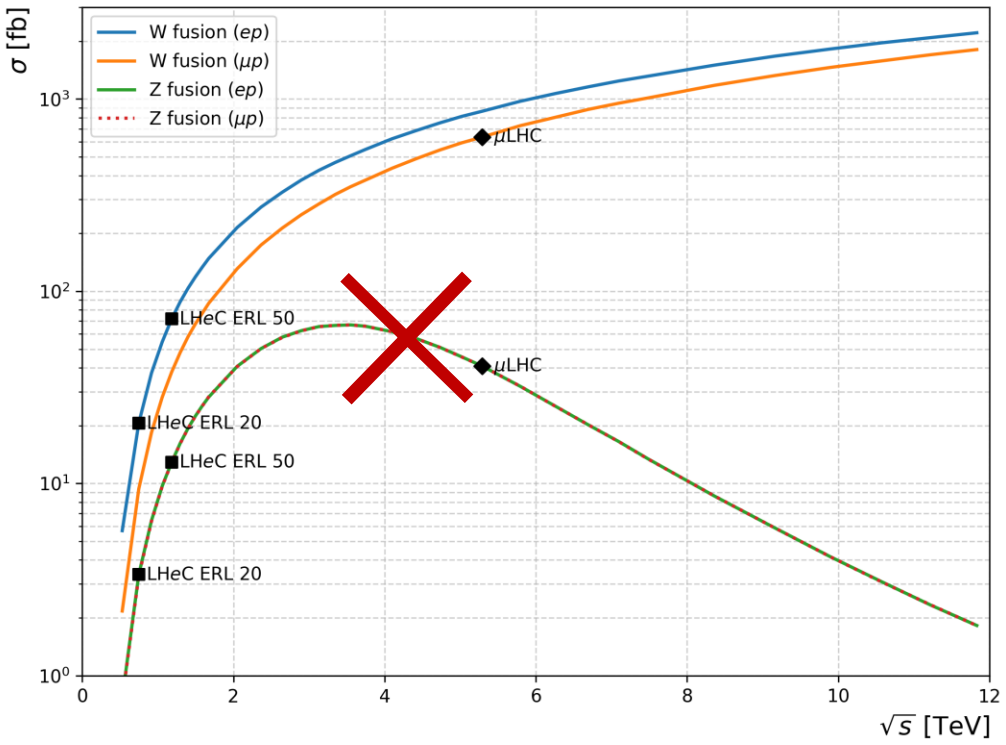
Higgs Production – MG5 vs CompHEP/CalcHEP

run_card.dat

```
#*********************************************************************
# BW cutoff (M+/-bwcutoff*Gamma) ! Define on/off-shell for "$" and decay
#*********************************************************************
 15.0  = bwcutoff      ! (M+/-bwcutoff*Gamma)
#*********************************************************************
# Standard Cuts                                                      *
#*********************************************************************
# Minimum and maximum pt's (for max, -1 means no cut)                *
#*********************************************************************
 20.0  = ptj       ! minimum pt for the jets
 10.0  = ptl       ! minimum pt for the charged leptons
 -1.0  = ptjmax    ! maximum pt for the jets
 -1.0  = ptlmax    ! maximum pt for the charged leptons
 {} = pt_min_pdg ! pt cut for other particles (use pdg code). Applied on
 {}     = pt_max_pdg ! pt cut for other particles (syntax e.g. {6: 100,
#
# For display option for energy cut in the partonic center of mass frame
#
#*********************************************************************
# Maximum and minimum absolute rapidity (for max, -1 means no cut)   *
#*********************************************************************
 5.0 = etaj    ! max rap for the jets
 2.5  = etal    ! max rap for the charged leptons
 0.0  = etalmin ! min rap for the charged leptons
 {} = eta_min_pdg ! rap cut for other particles (use pdg code). Applied
 {} = eta_max_pdg ! rap cut for other particles (syntax e.g. {6: 2.5, 23
#*********************************************************************
# Minimum and maximum DeltaR distance                                *
#*********************************************************************
 0.4 = drjl    ! min distance between jet and lepton
 -1.0  = drjlmax ! max distance between jet and lepton
#*********************************************************************
```

- MadGraph applies the cuts inside the matrix-element phase-space integral, so the result you get is a **fiducial (cut-dependent) cross section**, not the total one.
- **bwcutoff** also restricts the allowed invariant-mass window around resonances, so it further reduces the phase space.

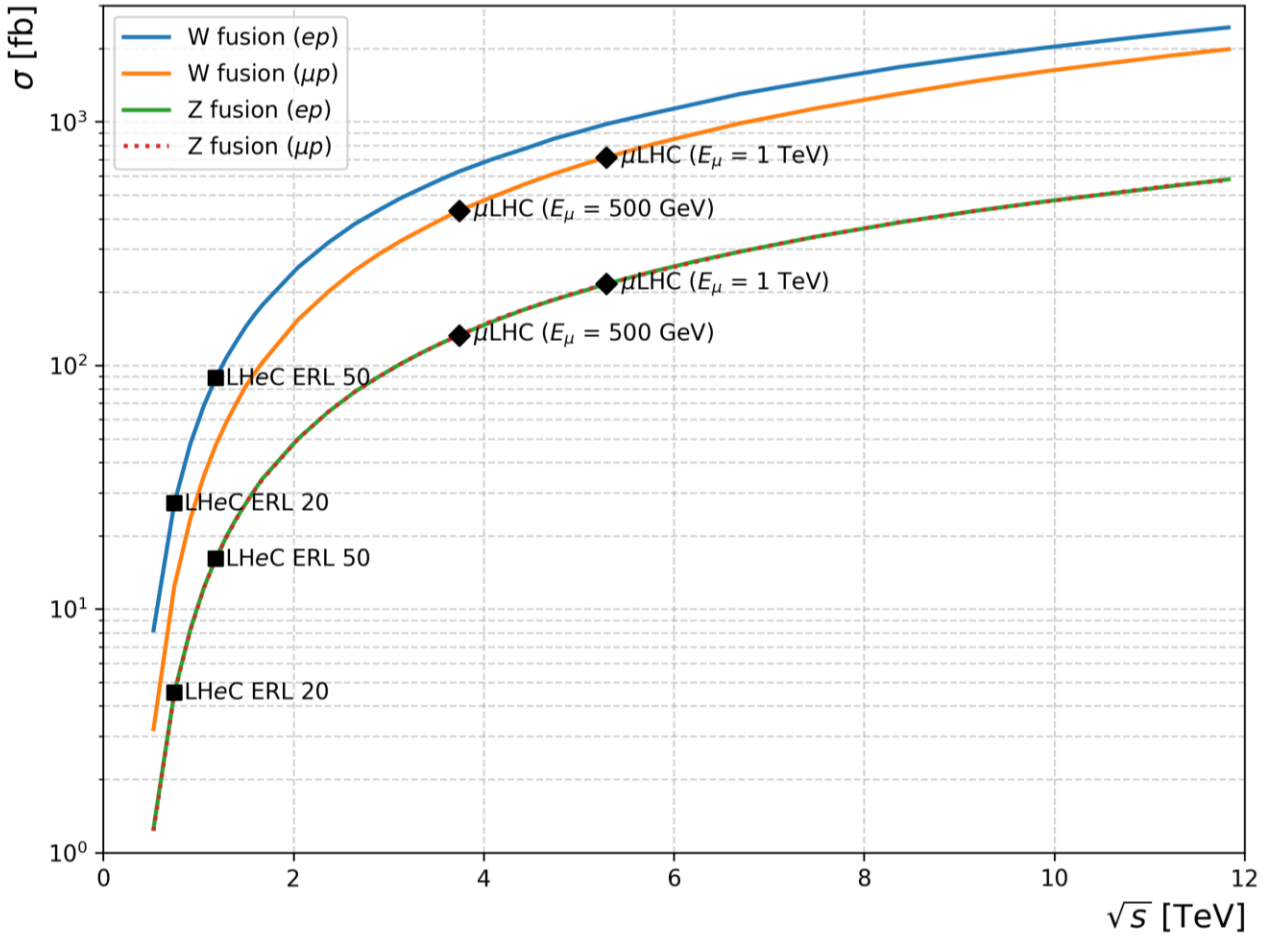


Higgs Production

Collider	$\sqrt{s}$ $[TeV]$	$L[cm^{-2}s^{-1}]$	$L_{int}[fb^{-1}/\text{year}]$	σ_{WWH} $[pb]$	$N_{Wfusion}/\text{year}$	σ_{ZZH} $[pb]$	$N_{Zfusion}/\text{year}$
LHeC (ERL 20)	0.748	10^{33} (10^{34})	10 (100)	0.027	270 (2700)	0.005	50 (500)
LHeC (ERL 50)	1.18	10^{34} (10^{35})	100 (1000)	0.089	8900 (89000)	0.016	1600 (16000)
μLHC (E_μ = 500 GeV)	3.7	10^{33} (10^{34})	10 (100)	0.432	4320 (43200)	0.133	1330 (13300)
μLHC (E_μ = 1 TeV)	5.3	10^{33} (10^{34})	10 (100)	0.714	7140 (71400)	0.216	2160 (21600)

		MG5 input
ep	W fusion	e- p > ve h j **/ j e-**
	Z fusion	e- p > e- h j **/ j e- h**
μp	W fusion	mu+ p > vm~ h j **/ j mu+**
	Z fusion	mu+ p > mu+ h j **/ j mu+ h**

```
set bwcutoff -1
set ptj -1
set ptl -1
set etaj -1
set etal -1
set drjl -1
```



Top Production

Process	cs [pb] default run card	cs [pb] op. 1	cs [pb] op. 2
$\mu^+ p \to \mu^+ t \bar{t}$	0.30	1.48	2.09
$\mu^+ p \to \mu^+ t \bar{t} X$	0.21	4.48	160.23
$\mu^+ p \to \mu^+ t X$	29.26	29.37	29.62

```
set ptj 1.0
set ptl 1.0
set etaj 9.0
set etal 9.0
```

```
set bwcutoff -1
set ptj -1
set ptl -1
set etaj -1
set etal -1
set drjl -1
```

- **MadGraph's default run card** applies built-in phase-space restrictions, just as seen in Higgs production studies, so the resulting cross section is cut-dependent rather than fully inclusive.
- **The first option** only relaxes these restrictions, which means it still yields a fiducial (cut-dependent) cross section.
- **The second option** removes all restrictions entirely, giving a result that is much closer to the fully inclusive LO theoretical cross section.

Which setup should be used in the analyses?